particle. The light quantum can behave as a particle and this is called photon. Therefore, photon is nothing but particle manifestation of light.

Characteristics of photons:

According to particle nature of light, photons are the basic constituents of any radiation and possess the following characteristic properties:

i) The photons of light of frequency v and wavelength λ will have energy, given by

$$E = hv = \frac{hc}{\lambda}.$$

ii) The energy of a photon is determined by the frequency of the radiation and not by its intensity and the intensity has no relation with the energy of the individual photons in the beam.

iii) The photons travel with the speed of light and its momentum is given by

$$p = \frac{h}{\lambda} = \frac{hv}{c}$$

iv) Since photons are electrically neutral, they are unaffected by electric and magnetic fields.

v) When a photon interacts with matter (photon-electron collision), the total energy, total linear momentum and angular momentum are conserved. Since photon may be absorbed or a new photon may be produced in such interactions, the number of photons may not be conserved.

Note

According to quantum concept, intensity of light of given wavelength is defined as the number of energy quanta or photons incident per unit area per unit time, with each photon having same energy. Its unit is Wm^{-2}.

Einstein's explanation of photoelectric equation

When a photon of energy hv is incident on a metal surface, it is completely absorbed by a single electron and the electron is ejected. In this process, a part of the photon energy is used in overcoming the potential barrier of the metal surface (photoelectric work function ϕ_0)and the remaining energy as the kinetic energy of the ejected electron. From the law of conservation of energy,

$$hv = \phi_0 + \frac{1}{2}mv^2 \qquad (8.6)$$

where m is the mass of the electron and v its velocity. This is shown in Figure 8.13(a).

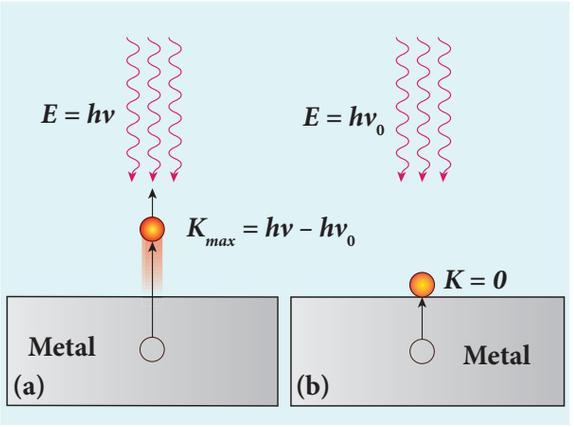


Figure 8.13 Emission of photoelectrons

If we reduce the frequency of the incident light, the speed or kinetic energy of photo electrons is also reduced. At some frequency v_0 of incident radiation, the photo electrons are just ejected with almost zero kinetic energy (Figure 8.13(b)). Then the equation (8.6) becomes

$$hv_0 = \phi_0$$

where v_0 is the threshold frequency. By

UNIT-8(XII-Physics_Vol-2)_New.indd 119 21-07-2020 16:42:00